

Macroeconomics I

Problem Set 5

1. **(Intergenerational Altruism).** Consider an OLG model where a mass of individuals with unit measure live two periods: childhood and adulthood. Individuals have warm-glow preferences, i.e., they value the bequests left for their children in their utility. In adulthood, the individual receives the inheritance from their parents (and rents it out as capital to firms), has children, works, chooses the inheritance for their children, and dies. The utility in childhood is not relevant (imagine that children's consumption is already incorporated into parents' consumption). The utility of an individual i reaching adulthood at t is:

$$\ln(c_t^i) + \beta \ln(b_t^i),$$

where b_t^i is the bequest left for their children and $\beta \in (0, 1)$. The budget constraint is

$$c_t^i + b_t^i = w_t + (1 + r_t - \delta)b_{t-1}^i,$$

and $b_0^i = b_0 > 0$ given.¹ The production side is standard: $Y_t = K_t^\alpha L_t^{1-\alpha}$ with $\alpha \in (0, 1)$. There is no population growth, and $L_0 = 1$.

- (a) Characterize the equilibrium of the economy. That is, solve the household's problem, the firm's problem, and write down the set of equations characterizing the optimal allocations and prices. How does capital evolve in this economy?
- (b) Find the steady state of k_{ss} , c_{ss} , b_{ss} , and y_{ss} as functions of the parameters.
- (c) Now suppose the individual values the TOTAL utility of their children (not just the bequest left). The utility of an individual reaching adulthood at t is:

$$U_t = \ln(c_t) + \beta U_{t+1}.$$

Everything else remains the same (add a no-Ponzi and a TVC). Characterize the equilibrium of the economy.

- (d) Suppose full depreciation, $\delta = 1$ (to simplify). Compare the steady-state capital in the two cases. Under what conditions are they equal? Explain intuitively how your answer would change if we used a CRRA function.
2. **(OLG with Distortionary Taxes).** Consider the standard OLG model where the agent works when young and consumes savings when old. All assumptions are usual and follow the model seen in class. The utility is:

$$\ln(c_t^1) + \beta \ln(c_{t+1}^2).$$

¹Note that b_0 is the same for all individuals, and you can ignore the index i . See Acemoglu's book for the case where individuals start with heterogeneous inheritance.

The agent's income is subject to two types of taxes: on labor τ_w and on capital τ_k . Suppose the tax revenue is thrown into the ocean. The agent's budget constraints when young and old are:

$$\begin{aligned} c_t^1 + s_t &\leq w_t(1 - \tau_w) \\ c_{t+1}^2 &\leq [1 + (r_{t+1} - \delta)(1 - \tau_k)]s_t. \end{aligned}$$

The production function is: $Y_t = K_t^\alpha L_t^{1-\alpha}$, with $\alpha \in (0, 1)$ and the population grows $L_t = (1 + n)^t L_0$ where $L_0 = 1$. The initial old generation has $k_0 > 0$ given.

- (a) Solve the household's problem. Find c_t^1 , c_{t+1}^2 , and s_t . Why does savings not depend on τ_k ?
 - (b) Use the solution to the firm's problem (standard) and the equilibrium condition in the asset market to find a first-order difference equation characterizing the optimal sequence of capital.
 - (c) Write down the equation determining capital in the steady state. How does it depend on τ_w ? Explain intuitively why the labor tax has a different effect compared to the standard neoclassical growth model (without leisure in the utility function and in infinite horizon).
3. **(OLG with Government).** Consider the standard OLG model where the agent works when young and consumes savings when old. The utility of an agent born in t is:

$$\ln(c_t^1) + \beta \ln(c_{t+1}^2).$$

The government taxes the young via lump-sum taxation, τ_t . Suppose $\delta = 0$. The budget constraints of the agent when young and old for all generations $t \geq 1$ are:

$$\begin{aligned} c_t^1 + s_t &\leq w_t - \tau_t \\ c_{t+1}^2 &\leq (1 + r_{t+1})s_t \end{aligned}$$

The government uses taxation to finance its fiscal policy. Suppose the budget is balanced every period, i.e., the government's budget constraint for all $t \geq 1$ is:

$$\tau_t L_t = G_t \equiv g_t L_t,$$

where $G_t > 0$ is the aggregate spending (and is exogenous), and g_t is spending per effective unit of labor. The production function is: $Y_t = K_t^\alpha L_t^{1-\alpha}$, with $\alpha \in (0, 1)$ and the population grows $L_t = (1 + n)^t L_0$ where $L_0 = 1$. The initial old generation has $K_0 > 0$ given.

- (a) Solve the households' problem. Find s_t in terms of the parameters, prices, and tax τ_t .
- (b) Write down the equilibrium conditions in the goods market and in the asset market.
- (c) Suppose the prices are given by $r_t = \alpha k_t^{\alpha-1}$ and $w_t = (1 - \alpha)k_t^\alpha$ for all t , where $k_t \equiv K_t/L_t$. Find an equation that describes the evolution of the equilibrium capital in the model (k_{t+1} in terms of k_t and parameters). How does an increase in government spending per worker alter the capital accumulation dynamics?

- (d) Suppose $n = 0$ (for simplicity). In the discrete-time neoclassical growth model with infinitely-lived dynasties and fiscal policy financed by lump-sum taxation, the Euler equation is given by:

$$c_{t+1} = c_t[\beta(1 + r_{t+1})] \quad \forall t.$$

Find the k_t in the steady state in the neoclassical growth model. How does the level of government spending per effective unit of labor, g , affect capital in the steady state in the neoclassical growth model? Compare with the OLG model.

4. (Families, Endogenous Fertility, and Human Capital).²

- (a) *Barro-Becker Endogenous Fertility Model* with human capital. A (uniparental) household derives utility from consumption (c), their number of children (n), and the future income of their children (y'):

$$\ln(c) + \gamma_n \ln(n) + \gamma \ln(y'). \quad (1)$$

Denote the household's human capital as H and the time spent on production as ℓ . The production of consumption is $c = y\ell$, where $y = AH$ is the total income of the household if they worked full-time.

The household's time is allocated to production and raising children. Denote e as the education given to the children and ϕ a fixed cost (in time) in raising children. The family's time constraint is:

$$\ell + n(\phi + e) \leq 1. \quad (2)$$

The human capital of children evolves according to the parents' human capital and the invested education: $H' = (Be)^\theta H$. Suppose that: $\gamma_n > \gamma\theta$.

- i. Show that the equilibrium education and fertility are:

$$e^* = \frac{\phi\gamma\theta}{\gamma_n - \gamma\theta} \quad \text{and} \quad n^* = \frac{\gamma_n - \gamma\theta}{\phi(1 + \gamma_n)}. \quad (3)$$

- ii. What is the (per-capita) growth rate of the economy (in terms of parameters)? What is the model's prediction for growth and fertility when there is a gradual increase in the return on human capital (θ)? What is the impact of a fertility restriction policy (like China's one-child policy) on per-capita economic growth?
- (b) **Bonus (not required to submit)** Bargaining power in *2-parent* households. Now consider a household consisting of a husband, a wife, a son, and a daughter (we will ignore fertility decisions). Households share consumption, but men and women disagree on how much they care about the welfare of their children. The household's utility is the sum of the parents' utility:

$$\lambda_f[\ln(c) + \gamma_f \ln(y')] + (1 - \lambda_f)[\ln(c) + \gamma_m \ln(y')], \quad (4)$$

²Question based on the chapter from Doepke and Tertilt (2016, Handbook of Macroeconomics): *Families in Macroeconomics*.

where λ_f is the woman's bargaining power in family decisions, γ_f and γ_m are the parameters of altruism of the woman and the man.

To simplify, consider that only women raise children. Women's time constraint is:

$$\ell + e_f + e_m \leq 1,$$

where e_f and e_m are the education invested in daughters and sons. Women and men are imperfect substitutes in the production of goods and human capital:

$$c = A(\ell H_f)^\alpha H_m^{(1-\alpha)}, \quad H'_f = (B e_f)^\theta H_f^\beta H_m^{1-\beta}, \quad \text{and} \quad H'_m = (B e_m)^\theta H_f^\beta H_m^{1-\beta},$$

and the total income if the woman works full-time is: $y = A(H_f)^\alpha H_m^{(1-\alpha)}$.

- i. Show that the equilibrium investment in education is:

$$e_f^* = \frac{\theta \alpha \delta}{\alpha + \delta \theta} \quad \text{and} \quad e_m^* = \frac{\theta(1 - \alpha) \delta}{\alpha + \delta \theta},$$

where $\delta \equiv \lambda_f \gamma_f + (1 - \lambda_f) \gamma_m$.

How does an increase in the relative productivity of women (α) change the gender education gap (i.e., e_f/e_m) and the time spent on children?

- ii. Derive the per-capita growth rate of this economy (y'/y). Suppose that women are more altruistic towards children: $\gamma_f > \gamma_m$. How does an increase in women's bargaining power λ_f alter the growth rate of this economy?